

Putting names to faces

Kathryn H. McWeeny, Andrew W. Young, Dennis C. Hay and Andrew W. Ellis

Subjects were asked to learn occupations and surnames belonging to 16 unfamiliar faces. The experiment was intended to remove potential factors that might underlie the easier recall of people's occupations found in everyday life. Thus, contextual cues were eliminated by presenting faces in isolation, cueing from the retrieval of related items of semantic information was eliminated, and the frequency of use of occupational and name information was equated. Results showed that names remained much harder to recall than occupations. This was true even for ambiguous labels that could be used as names or as occupations. It is much harder to recall that a person's surname is Baker than to recall that a person is a baker.

A common experience is to know exactly who a seen person is but be unable to bring to mind her or his name. When in this state we usually know the person's occupation, where she or he is usually seen, who her or his friends are, and so on; only the name remains elusive (Young *et al.*, 1985). In contrast to these recall difficulties, recognition memory for names seems to be highly effective (Clarke, 1934; Bahrck *et al.*, 1975).

Problems in recalling people's names produce tip-of-the-tongue states which have been studied in order to further understanding of the processes involved in the retrieval of items from memory (Gruneberg *et al.*, 1973; Yarmey, 1973; Williams & Santos-Williams, 1980; Williams & Hollan, 1981; Read & Bruce, 1982; Reason & Mycielska, 1982; Reason & Lucas, 1984). What has received less attention, however, is the question as to why people's names should be so difficult to recall (though a notable exception is the work of Morton *et al.*, 1985).

The answer to this question is by no means obvious. Only a limited range of semantic characteristics, such as age or sex, can be reasonably accurately inferred from a person's facial appearance. Such characteristics as a person's occupation or the place where she or he is usually encountered are not closely linked to particular face shapes or certain types of facial feature. People do, of course, have occupational stereotypes concerning what, say, a 'typical' film star or a 'typical' pop star might look like (Klatzky *et al.*, 1982*a, b*), but these do not provide veridical information. Thus, in the absence of contextual cues, knowing a seen person's occupation can be as much a recall task as knowing that person's name.

Despite this point, Hay & Young (1982) and Young *et al.* (1985) noted that whereas we can often remember a seen person's occupation without being able to remember her or his name, we never seem to remember a seen person's name without also remembering their occupation. Hay & Young (1982), H. D. Ellis (1986), Young *et al.* (1985) and Bruce & Young (1986) used these, and other, observations to suggest that names are only accessed from familiar faces via intervening semantic codes that specify who the people are.

There is, then, a *prima facie* case for regarding names as relatively difficult things to recall as compared to other semantic information. Our interest in this paper is with why they are hard to recall. Although it has been argued that names are relatively difficult to recall because they can only be accessed indirectly from familiar faces via intervening semantic codes (Hay & Young, 1982; Young *et al.*, 1985; Bruce & Young, 1986; H. D. Ellis, 1986), it is easy to think of more prosaic potential explanations of the phenomenon.

The most obvious possibility is that everyday life simply provides us with a wide variety of non-facial contextual cues to occupation and the like. Thus we might know that Ms A. is a doctor because she is carrying a stethoscope, we might know that Mr B. works for the railways because he is checking our tickets, and so on. The errors studied by Young *et al.* (1985) all came from descriptions of real-life incidents, in which the role of such contextual cues would be difficult to assess.

It is also likely that, although we often need to recall them, names do not have to be retrieved with anything like the frequency that other types of semantic information are required. In addition, the retrieval of some items of semantic information may then cue the retrieval of others. Thus, if you remember that a person you meet in town is someone you usually see in a field it will be easier to remember that she or he is a local farmer.

Last, but by no means least, comes the point that names are often items of low meaningfulness or imageability (Cohen, Higgins, Jones, etc.) and therefore possibly unmemorable for this reason alone.

Thus simple explanations as to why people's names are retrieved less easily than other semantic information can be based on contextual cues, frequency of use, cues from other semantic information, and low meaningfulness or imageability. All of these explanations imply that if such factors could be equated between names and other items of semantic information, then names would be no more difficult to retrieve than anything else. In contrast, the explanation favoured by Young *et al.* (1985) and Bruce & Young (1986), that names are only retrieved via intervening semantic codes specifying a person's identity, would predict that a difference between ease of recall of names and other semantic information will continue to exist even when differences in cueing, frequency, and meaningfulness or imageability have been removed.

The experiment reported here tests these contrasting predictions. The factors underlying the simple explanations of difficulties in name retrieval were eliminated by showing subjects faces of unfamiliar people, and teaching them their (invented) names and occupations. Thus we could eliminate contextual cues by presenting each face in isolation, we could control the frequency of use of semantic or name information, and we could eliminate the role of cueing from other items of semantic information by teaching subjects only an occupation and a name for each person. To examine the effect of meaningfulness or imageability of individual items we made use of what we will call ambiguous labels; these have the property that they can be used as surnames or as occupations (Carpenter, Farmer, etc.).

Method

Subjects

Sixteen university students (eight male, eight female) acted as subjects. All were paid.

Stimuli and procedure

Black-and-white photographs of the faces of 16 middle-aged men who were unfamiliar to the subjects were used as stimuli. These photographs were surrounded by a closely fitting circular template to remove as much clothing and background as possible, and rephotographed to a standard size to make the slides used in the experiment.

Subjects were tested individually, and given spoken instructions by the experimenter. The 16 faces were shown to each subject one at a time, and s/he was told the person's (invented) surname and occupation. Half the subjects of each sex were told this information in the order surname then occupation, and half in the order occupation then surname. Thus half the subjects were told that 'This man is called Mr X and he is a Y', and half were told that 'This man is a Y and he is called Mr X'. Subjects were asked to try to remember each person's occupation and surname.

Each subject was then shown the 16 faces again, and was asked to state the person's occupation

and surname in the order in which s/he had learnt them. Any errors were noted by the experimenter, and corrected. This procedure was repeated, with 16 faces being presented in an unpredictable sequence for each repetition, until each subject had given correct occupations and surnames to all of the faces on two successive occasions.

Surnames to be paired with the faces were chosen by searching the Lancaster, Morecambe and Heysham March 1984 Telephone Directory for surnames which could be used as either surnames or as occupations, such as Baker, Cook or Porter. These names which can be used either as surnames or occupations are referred to here as ambiguous labels. Sixteen of these ambiguous labels were chosen. There were then divided into two sets of eight ambiguous labels which were matched as closely as possible on frequency of occurrence as a surname in the Lancaster Telephone Directory, on frequency of occurrence as a word in the Kucera & Francis (1967) norms, and for the number of letters in the labels.

A further eight surnames which could not be used as occupations (such as Hyde, Knowles, Rothwell) were then chosen to match the ambiguous labels on frequency of occurrence in the Lancaster Telephone Directory and for the number of letters in the items. We refer to these as unambiguous surnames.

Finally, eight occupations which are not commonly used as surnames were chosen (architect, grocer, solicitor, etc.). These unambiguous occupations were matched to the ambiguous labels on word frequency using the Kucera & Francis (1967) norms, and on number of letters.

The characteristics of these matched sets of items are given in Table 1.

Table 1. Characteristics of matched sets of items used in the experiment

	Description	Mean Telephone Directory frequency	Mean Kucera- Francis frequency	Mean number of letters
Set A	Ambiguous labels	13.50	14.25	6.38
Set B	Ambiguous labels	13.75	13.00	6.13
Set C	Unambiguous surnames	13.75	—	6.00
Set D	Unambiguous occupations	—	13.25	6.88

These sets of items were arranged into two groups of stimuli. The first group used one of the sets of ambiguous labels as surnames coupled to the unambiguous occupations, and the unambiguous surnames with the other set of ambiguous labels as occupations. In the second group of stimuli use of the two sets of ambiguous labels as occupations or as surnames was reversed. Thus the same ambiguous label did not appear as both surname and occupation in the same group of stimuli, and each pairing of surname and occupation only contained one ambiguous label. Use of these two groups of stimuli was counterbalanced across subjects in order to ensure that as many subjects learnt each ambiguous label as a surname as learnt that label as an occupation.

Thus subjects learnt occupations and surnames belonging to 16 faces, to a criterion of two correct runs through the entire 16 faces. The occupations that they learnt could be unambiguous (i.e. occupations that are not usually surnames) or ambiguous (occupations that could be surnames), and the surnames could also be unambiguous (i.e. surnames that are not occupations) or ambiguous (surnames that could be occupations). Errors made while learning these occupations and surnames were recorded.

Results

On each experimental trial the subject was presented with a face and asked to give the person's occupation and surname. On average, subjects needed to go through the sets of 16 faces 7.38 times (SD 1.41) in order to reach our criterion of two entirely correct runs through the 16 faces.

The responses for each trial can effectively be divided into four types: those in which both occupation and surname were correctly given, those in which only the occupation was

correctly given, those in which only the surname was correctly given, and those in which neither item was correctly given. The mean number of trials on which both occupation and surname were correctly given was 66.06 (SD 11.49), and the mean number of trials on which neither item was correctly given was 24.94 (SD 10.24). However, it is the numbers of trials in which only the occupation was correctly given and the numbers of trials in which only the name was correctly given that are of most interest.

The numbers of occasions on which only the occupation was correctly given or only the name was correctly given were totalled for each subject. These totals were then subdivided into those in which the unrecalled item belonged to one of the ambiguous or unambiguous sets. Means and standard deviations are presented in Table 2, where the data are also subdivided according to whether subjects were made to learn and recall the items in the order name then occupation or in the order occupation then name.

Table 2. Means and standard deviations of numbers of trials on which subjects recalled only an occupation or only a name

		Occupation recalled		Name recalled	
		Unrecalled ambiguous name	Unrecalled unambiguous name	Unrecalled ambiguous occupation	Unrecalled unambiguous occupation
Name then occupation	$\bar{X}$	11.63	9.25	0.38	0.63
	SD	6.16	5.68	0.74	0.74
Occupation then name	$\bar{X}$	15.25	12.13	1.13	1.13
	SD	7.69	2.90	2.03	1.81
Overall	$\bar{X}$	13.44	10.69	0.76	0.88
	SD	6.99	4.60	1.53	1.36

The most striking feature of the data was that every subject recalled an occupation without a name more often than s/he recalled a name without an occupation; this observation held for both the ambiguous and the unambiguous occupations and names. In fact the occasions on which a name was remembered without an occupation were so rare that the data in these cells of the design showed little variance. An $F_{\max}$ test applied to the data as presented in Table 2, following the procedure recommended by Winer (1962), showed that the assumption of homogeneity of variance necessary for an analysis of variance had been violated ($F_{\max} = 107.27$, $k = 8$, $n = 7$, $P < 0.001$). For this reason the data were subjected to a square root transformation. The effect of this transform was to reduce variances in the cells containing the larger scores, and thus satisfy the assumption of homogeneity of variance ($F_{\max} = 2.90$, $k = 8$, $n = 7$, $P > 0.1$).

A three-factor analysis of variance of these transformed data was carried out to determine the effects of order of learning (name then occupation, or occupation then name), item recalled (occupation recalled or name recalled; repeated measure), and item ambiguity (ambiguous or unambiguous; repeated measure). This revealed only one statistically significant effect, in which occupations were recalled more often than names (item recalled, $F = 139.88$, d.f. = 1, 14, $P < 0.001$).

The results of the experiment thus show that surnames are harder to recall than occupations, regardless of the order in which the items are learnt and regardless of their ambiguity or lack of ambiguity.

In order to check that the effect was not simply due to a few surnames being particularly difficult to recall, an items analysis was also carried out. The use of two groups of stimuli meant that a total of 32 name+occupation pairings were used in the experiment. For 31 of these pairings the name was the more frequently forgotten item (binomial test, $z = 5.13$, $P < 0.001$). Thus it is clear that the finding that surnames are harder to recall than occupations is of general applicability.

Discussion

The experiment was designed to eliminate factors that might underlie the relative difficulty of retrieving names as compared to other semantic information. Thus contextual cues to occupation were eliminated by presenting the faces in isolation, cueing from the retrieval of related items of semantic information was eliminated by only giving subjects one piece of semantic information (the person's occupation), and frequency of use of semantic or name information was controlled by asking subjects to give the person's name and occupation on every trial. In addition, ambiguous items were used as names for some subjects or as occupations for others.

The results of the experiment are unequivocal. Even when all of the factors mentioned have been controlled, people's names are much harder to recall than their occupations. This finding holds regardless of the order in which subjects learn and produce the items (name then occupation, or occupation then name), and it holds both for ambiguous and for unambiguous items. It is much more difficult to recall that someone's name is *Mr Potter* than to recall that he is *a potter*.

This finding that people's names are intrinsically harder to recall than other types of semantic information is predicted by the view taken by Young *et al.* (1985) and Bruce & Young (1986), who argue that as we learn about familiar people we create 'person identity nodes' which hold essential semantic codes and that names are then accessed via these person identity nodes. The purpose of such identity nodes is to hold the semantic codes that specify a person's identity, and that need to be accessed from different recognition systems (face, voice, build, etc.). These identity nodes thus form the point at which person recognition (rather than face recognition, voice recognition, etc.) is achieved. Bruce & Young (1986) discuss a range of evidence supporting this view, including the fact that it takes people longer to name faces than to categorize them by occupation (Young *et al.*, 1986) and a report of a clinical case involving an anomia restricted to proper names (McKenna & Warrington, 1980).

The results of the experiment show, however, that errors can occur in which a person's name is recalled without correct recall of his or her occupation. These have not been found for errors made in everyday life (Young *et al.*, 1985). The design of the present experiment should, of course, involve conditions that are ideal for the production of such errors. Nonetheless, they are by no means frequent, but the fact that they can occur at all shows that it is not in principle impossible for people to attach names directly to faces. Thus the evidence that people's names are usually accessed via intervening semantic information (Young *et al.*, 1985; Bruce & Young, 1986) cannot be attributed to a complete inability to form direct face to name links.

Although the results of our experiment are predicted by the Bruce & Young (1986) model, which thus gains further empirical support, they are not inconsistent with *all* alternative theoretical positions. We sought to falsify as many alternative explanations as could be achieved in a single experiment, but others of course remain to be subjected to further tests. The most notable of these is the possibility that subjects are more likely to use imagery techniques to assist in linking occupations and faces than they are to use such techniques in linking names and faces. Thus they might visualize the potter at work, or

think that his face looks like or unlike a typical potter, whereas they would perhaps not do the same for the name Mr Potter. Indeed, an effective mnemonic for improving ability to learn people's names makes use of imagery (Lorayne, 1958; Morris *et al.*, 1978), which does suggest that this strategy is not usually used spontaneously. In fact, when asked how they learnt the names and occupations, a number of subjects (but not all) commented that they made at least some use of occupational stereotypes in learning the occupations. Thus they might maintain that the baker's face looked like a baker, whereas they did not use such strategies with the names.

The results of our experiment are thus consistent with the idea that names are learned in a quite different way to other semantic information and, as we have noted, further studies should be able to throw more light on what is happening. We would suggest that it might be the case that any incidental meanings names can have are ignored, so that they are learned as purely arbitrary labels. In the 20th century we do not, for instance, want to think that Ms Carpenter might really be a carpenter. As anecdotal evidence for this idea we offer the observations that we found it very difficult not to overlook many of the ambiguous names when searching for them in the local Telephone Directory, and we subsequently began to realize that acquaintances possessed names of this type that we had not noticed before.

An alternative way of thinking about the problems that people experience in remembering names has been put forward by Morton *et al.* (1985). They suggest that information is held in memory units, each of which can only be accessed by a particular key, and use the term 'records' to describe the memory units and the term 'headings' to describe the access keys. Morton *et al.* (1985) propose that any type of information may be contained in a record and any type of information may function as a heading, but that the heading is represented in a different form to the record. Although the heading is used to access the record, the content of the heading itself cannot be retrieved. Thus Morton *et al.* (1985) explain that the name 'Bill Smith' might function as the heading for a record containing the information as to who Bill Smith is. If this particular record were only to contain semantic information about Bill Smith, and did not itself specify that his name is 'Bill Smith', then the name would be unrecalled because it is in the inaccessible heading. The headed records theory allows that different and independent records may contain information about Bill Smith, so that in other circumstances a record containing his name might well be found.

The headed records model (Morton *et al.*, 1985) provides a particularly neat account of the problems people experience in remembering names under everyday conditions. We are not certain, however, that it accounts readily for the present data. According to the headed records model a record is either accessed or not accessed; there is no 'partial retrieval'. In the present experiment, subjects are initially told and then either recall or are reminded of both a name and an occupation for the face presented on each trial. Thus the greater difficulty of remembering names would presumably arise from subjects tending to create records that contained occupations but not names. Although it is possible that this is what happens, and a good reason as to why it happens may be forthcoming, such a phenomenon does not seem to be predicted by the headed records model as it stands.

We have shown, then, that the reasons why names are harder to recall than occupations are more interesting and less obvious than they might at first seem. Most likely they derive from a change in what people actually do with an item according to whether it is a name or an occupation. The phenomenon of difficulty in recalling names is often remarked upon in everyday life, and as such it is something that psychologists need to be able to explain. Attempting to account for such everyday phenomena helps to ensure that the findings of

our studies can be of use outside the laboratories in which they are usually conducted (Baddeley, 1981; A. W. Ellis, 1983).

Acknowledgements

This research was supported by the ESRC (Grant C 0023 2075). We are grateful to Peter Morris and Bill Phillips for insisting on the importance of simple explanations of difficulties in remembering names, to Hadyn Ellis and Jim Reason for helpful comments on previous drafts of the paper, and to Vicki Bruce and Mike Burton for drawing our attention to the usefulness of names that can also be occupations.

References

- Baddeley, A. (1981). The cognitive psychology of everyday life. *British Journal of Psychology*, **72**, 257–269.
- Bahrick, H. P., Bahrick, P. O. & Wittlinger, R. P. (1975). Fifty years of memory for names and faces: A cross-sectional approach. *Journal of Experimental Psychology: General*, **104**, 54–75.
- Bruce, V. & Young, A. W. (1986). Understanding face recognition. *British Journal of Psychology*, **77**, 305–327.
- Clarke, H. M. (1934). Recall and recognition for faces and names. *Journal of Applied Psychology*, **18**, 757–763.
- Ellis, A. W. (1983). Syndromes, slips and structures. *Bulletin of The British Psychological Society*, **36**, 372–374.
- Ellis, H. D. (1986). Processes underlying face recognition. In R. Bruyer (ed.), *The Neuropsychology of Face Perception and Facial Expression*. Hillsdale, NJ: Erlbaum.
- Gruneberg, M. M., Smith, R. L. & Rinfrow, P. (1973). An investigation into response blocking. *Acta Psychologica*, **37**, 187–196.
- Hay, D. C. & Young, A. W. (1982). The human face. In A. W. Ellis (ed.), *Normality and Pathology in Cognitive Functions*. London: Academic Press.
- Klatzky, R. L., Martin, G. L. & Kane, R. A. (1982a). Influence of social-category activation on processing of visual information. *Social Cognition*, **1**, 95–109.
- Klatzky, R. L., Martin, G. L. & Kane, R. A. (1982b). Semantic interpretation effects on memory for faces. *Memory and Cognition*, **10**, 195–206.
- Kucera, H. & Francis, W. N. (1967). *Computational Analysis of Present-day American English*. Providence, RI: Brown University Press.
- Lorayne, H. (1958). *How to Develop a Super-Power Memory*. Preston: Thomas.
- McKenna, P. & Warrington, E. K. (1980). Testing for nominal dysphasia. *Journal of Neurology, Neurosurgery and Psychiatry*, **43**, 781–788.
- Morris, P. E., Jones, S. & Hampson, P. (1978). An imagery mnemonic for the learning of people's names. *British Journal of Psychology*, **69**, 335–336.
- Morton, J., Hammersley, R. H. & Bekerian, D. A. (1985). Headed records: A model for memory and its failures. *Cognition*, **20**, 1–23.
- Read, J. D. & Bruce, D. (1983). Longitudinal tracking of difficult memory retrievals. *Cognitive Psychology*, **14**, 280–300.
- Reason, J. T. & Lucas, D. (1984). Using cognitive diaries to investigate naturally occurring memory blocks. In J. Harris & P. E. Morris (eds), *Everyday Memory, Actions and Absentmindedness*. London: Academic Press.
- Reason, J. T. & Mycielska, K. (1982). *Absent Minded? The Psychology of Mental Lapses and Everyday Errors*. Englewood Cliffs, NJ: Prentice-Hall.
- Williams, M. D. & Hollan, J. D. (1981). The process of retrieval from very long-term memory. *Cognitive Science*, **5**, 87–119.
- Williams, M. D. & Santos-Williams, S. (1980). Method for exploring retrieval processes using verbal protocols. In R. S. Nickerson (ed.), *Attention and Performance*, vol. 8. Hillsdale, NJ: Erlbaum.
- Winer, B. J. (1962). *Statistical Principles in Experimental Design*. London: McGraw-Hill.
- Yarmey, A. D. (1973). I recognize your face but I can't remember your name: Further evidence on the tip-of-the-tongue phenomenon. *Memory and Cognition*, **3**, 287–290.
- Young, A. W., Hay, D. C. & Ellis, A. W. (1985). The faces that launched a thousand slips: Everyday difficulties and errors in recognizing people. *British Journal of Psychology*, **76**, 495–523.
- Young, A. W., McWeeny, K. H., Ellis, A. W. & Hay, D. C. (1986). Naming and categorizing faces and written names. *Quarterly Journal of Experimental Psychology*, **38A**, 297–318.

Received 22 November 1985; revised version received 1 May 1986

Requests for reprints should be addressed to Andy Young, Psychology Department, Lancaster University, Lancaster LA1 4YF, UK.